

BETA-HP-200 Hydraulic Press - Energy Control Program

Field	Value
Tenant	Beta Manufacturing (tenant-beta)
Site	Harbor Works, Line 1 (imperial documentation)
Document identifier	BETA-SAF-HP200-ECP-1
Source identifier	BM-DOC-2231
Equipment	BETA-HP-200
Also covers	BETA-HP-200S (additional requirements under Additional requirements for BETA-HP-200S)
Document type	Safety / energy isolation procedure
Revision	Rev 1
Effective date	2023-03-01
Status	current
Applies to	Assets listed under equipment
Related documents	BETA-MAN-HP200-B, BETA-PM-HP200-B, BETA-DIA-P100-1
Document owner	EHS and Maintenance Engineering

Purpose and scope

This program describes how electrical, hydraulic, die cushion and gravitational energy are made safe on the BETA-HP-200 press before any person opens a guard, enters the die area, touches a tie rod, handles a hose or fitting, or opens the manifold.

It applies to BETA-HP-200 and to BETA-HP-200S.

Rule 1. No person may open a guard, enter the die area or break a hydraulic line on this press unless this program has been completed for the specific task and the red verification tag is attached.

Rule 2. On this press a two-person rule applies. The person performing the work and a second authorised person who watches the energy control steps must both sign the verification tag. This rule is not a formality: the die cushion stores energy that is not visible from the operating station.

Energy sources on this machine

Energy	Source	Isolation point
Electrical, 460 V three phase	Feeder BM-MCC-3	Panel 3B-11, breaker 7
Electrical, control 24 V DC	Control transformer in panel 3B-11	Breaker 9
Hydraulic, main circuit	Power unit P-100, pumps A and B	Ball valves HX-301 and HX-302
Hydraulic, tool clamp circuit	Clamp manifold	Ball valve HX-311
Stored pressure, accumulator AC-2	Nitrogen pre-charge, 1000 psi as built	Manual bleed valve BV-321 into drain drum DD-2
Stored pressure, die cushion	Cushion cylinder	Cushion release valve RV-330, then cushion block CB-9
Gravitational	Ram mass, about 900 lb	Ram block RB-6 plus tie-off
Pneumatic, control only	Instrument air at 60 psi	Ball valve AV-315
Residual	Disconnected hoses	Cap on removal, blank plug before work

The die cushion is the difference between this press and the four-post presses elsewhere on the site. Blocking the ram alone does not make the cushion safe.

Required sequence

Perform these steps in order. Skipping a step is a reportable event.

1. Notify. Inform the Line 1 shift supervisor and the press operators that the press will be down, and record it on permit LOTO-4410.
2. Stop normally. Press stop at the operator station and allow the pumps to run down for the full 60 seconds. Do not use the emergency stop as a normal stopping method.
3. Isolate electrical energy. Open panel 3B-11, set breaker 7 to OFF and the isolator handle to the locked position, and fit a personal lock and tag. Isolate the control supply at breaker 9. Use the lock box when more than one person is working.
4. Isolate the hydraulic supply. Close HX-301 and HX-302, and lock both. Close HX-311 on the clamp circuit and lock it. Close instrument air at AV-315.
5. Dissipate the die cushion energy. Open cushion release valve RV-330 and let the cushion extend. Fit the cushion block CB-9 in the ram yoke before the ram is moved. The cushion pressure is only released when RV-330 is open and the ram yoke has dropped onto CB-9.
6. Dissipate the accumulator energy. Open bleed valve BV-321 and discharge accumulator AC-2 into drain drum DD-2. Never vent a charged accumulator to atmosphere. Continue bleeding until the accumulator gauge reads 10 psi or less. If the gauge will not come below 10 psi, stop and escalate; the accumulator or the gauge is defective and the circuit must be treated as charged.
7. Fit the ram block and tie-off. Fit ram block RB-6 under the ram yoke and fit the tie-off. Both must be fitted before hands enter the die area.
8. Verify the zero-energy state. Perform all five verifications:
 - Attempt start from the operator station: nothing moves.
 - Attempt to operate the locked ball valves: they do not move.
 - Attempt to move the ram by hand at the ram crown: it does not move.
 - Confirm the cushion pressure gauge at the cushion manifold reads 0 psi.
 - Confirm the accumulator gauge reads 10 psi or less and the main pressure gauge reads 0 psi.Attach the red verification tag. Both the technician and the second authorised person sign it.

The attempt start verification is recorded on permit LOTO-4410 and is audited monthly.

Rules that are frequently broken

Working on a hose or fitting because the pump motor is off. Motor off is not isolation.

Blocking the ram without releasing the cushion. The cushion can still drive the ram down.

One person signing the verification tag.

Assuming the accumulator is empty because it was bled on a previous job.

The gauge reading on the day, by that person, is the evidence.

Removing a hose and leaving the port open. Cap the port the same hour.

Leaving a personal lock off during a break. Every break means re-verification.

Bypassing the light curtain to run the press in setup mode. Prohibited.

Return to service

- 1. Remove tools, blanks and test pieces from the die area. Fit all guards.
- 2. Close BV-321, remove drain drum DD-2, close RV-330 after confirming the cushion is recharged on the next cycle.
- 3. Remove the cushion block CB-9, the ram block RB-6 and the tie-off and return them to the store.
- 4. Remove locks one at a time; each person removes their own lock.
- 5. Notify the Line 1 supervisor that the press is released.
- 6. Restore panel 3B-11 and close HX-301, HX-302, HX-311 and AV-315.
- 7. Recharge accumulator AC-2 through the charging point. The pre-charge value is the as-built value on the data plate. Do not charge from a figure copied from another press on this site.
- 8. Run a dry cycle with the die area empty and hands clear. Confirm the cushion recharges, the clamp pressure proves, the ram raises and lowers normally and there is no leak.
- 9. Record the return to service on permit LOTO-4410 and on the work order.

Additional requirements for BETA-HP-200S

The straight-side variant has a fixed guard and a two-tool clamp arrangement. Before work on its clamp circuit, both clamp circuits must be isolated and bled separately, because the two clamp circuits are cross connected upstream of HX-311.

Work instruction WI-2204 covers the HP-200S and is not part of this document collection. Where this program and WI-2204 differ, raise a documentation conflict; do not choose between them in the field.

Incident reporting

Report to EHS before the next shift, even if nobody was hurt:

- A guard was opened or a fitting loosened while the circuit was charged.
- A person entered the die area with the cushion charged.
- An attempt start produced any movement of the ram.
- A tag was found without a lock, or a lock was found unlocked.
- Work was performed without a two-person verification.
- A safety device was found bypassed, taped or missing.

Appendix: Energy isolation point schedule

The energy isolation points for the BETA-HP-200 press and its hydraulic power unit are listed below. Every point is locked and tagged before work begins. The schedule is verified against the as-built drawing at each annual review.

Ref	Energy source	Device	Location	Method	Verification
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HV-101	Electrical, main	Isolator	Panel 1A-02	Lock open	Test for dead at motor terminals
HV-201	Electrical, control	Isolator	Panel 1A-02	Lock open	Test control voltage absent
HY-1	Hydraulic, stored	Relief and drain valve	Manifold	Open and drain	Gauge reads zero

AC-1	Pneumatic, accumulator	Block valve BV-103	Manifold	Open to drain vessel	Accumulator gauge reads zero
ME-1	Mechanical, gravity	Ram block and tie-off	Frame	Fit mechanical stop	Ram cannot move under load
TH-1	Thermal	Oil cooler	Power unit	Allow to cool	Below 104 degF

Isolation is not complete until every point has been applied and the zero-energy state has been verified by attempting to start the press and confirming nothing moves. The attempt is recorded with date, time and name.

Two people are required: the person who applied the isolation and a second person who confirms the zero-energy state independently.

Appendix: Permit and record retention

The isolation permit is raised for each job and closed only after the isolation has been removed and the press returned to service. The permit identifies the asset, the work order, the isolation points applied, the verifying person and the time of removal.

Records are retained for seven years. The record set for each job contains:

the raised permit and the lock and tag numbers issued;
the zero-energy verification record with the attempted-start result;
the list of isolation points and how each was proven dead;
the name of the person who applied the isolation and the verifier;
the time the isolation was removed and the press returned to service;
any deviation and the approval that authorized it.

A permit that cannot be produced when requested is treated as an unverified isolation and the work is stopped until the record is reconstructed and re-verified.

Revision history

Revision	Effective date	Summary
Rev 1	2023-03-01	First controlled issue, issued with the energy control program audit.